

MH1100 Calculus I

Handout 13 - Inverse Function

Inverse Function

- Indeterminate Forms
- L'Hospital's Rule

Indeterminate Forms and L'Hospital's Rule

We would like to know the value of the limit

$$\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}. \quad \frac{0}{0}, \frac{\infty}{\infty} \rightarrow \text{Very hard if (odd)}$$

The limit is not obvious, because both numerator and denominator approach 0 and $\frac{0}{0}$ is not defined.

In general, if we have a limit of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)},$$

where both $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow a$, the limit may or may not exist. This form is called an **indeterminate form of type $\frac{0}{0}$** .

Indeterminate Forms and L'Hospital's Rule

Previously, we have two approaches as follows:

(1) For rational functions, we can cancel common factors:

$$\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x(x-1)}{(x+1)(x-1)} = \lim_{x \rightarrow 1} \frac{x}{x+1} = \frac{1}{2}.$$

(2) We used a geometric argument to show that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

But these methods do not work for the limit $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$.

Indeterminate Forms and L'Hospital's Rule

Moreover, when we calculate the horizontal asymptote for $F(x) = \frac{\ln x}{x-1}$, we need to know

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x-1} \quad (2).$$

It isn't obvious how to evaluate this limit, because both numerator and denominator become ∞ as $x \rightarrow \infty$.

In general, if we have a limit of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where both $f(x) \rightarrow \infty$ (or $-\infty$) and $g(x) \rightarrow \infty$ (or $-\infty$), the limit may or may not exist. This form is called an **indeterminate form of type $\frac{\infty}{\infty}$** .

Indeterminate Forms and L'Hospital's Rule

L'Hospital's Rule applies to these indeterminate forms.

L'Hospital's Rule

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \text{ and } \lim_{x \rightarrow a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or ∞/∞). Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

$f(a), g(a)$
 $f'(a), g'(a)$ $\rightarrow$ don't need to be well defined when using L'Hopital Rule

if the limit on the right side exists (or is ∞ or $-\infty$).

Indeterminate Forms and L'Hospital's Rule

Note 1: L'Hospital's Rule says that the limit of a quotient of functions is equal to the limit of the quotient of their derivatives. It is especially important to verify the conditions regarding the limits of $f(x)$ and $g(x)$ before using L'Hospital's Rule.

Note 2: L'Hospital's Rule is valid for **one-sided limits** and for **limits at infinity or negative infinity**. That is, " $x \rightarrow a$ " can be replaced by any of the symbols $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, or $x \rightarrow -\infty$.

Note 3: When $f(a) = g(a) = 0$, $f'(x)$ and $g'(x)$ are continuous, and $g'(a) \neq 0$, it is easy to see why L'Hospital's Rule is true. In fact, using the alternative form of the definition of a derivative, we have

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)} = \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}} = \lim_{x \rightarrow a} \frac{\cancel{f(x) - f(a)}}{\cancel{g(x) - g(a)}} = \lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

$f(a) = g(a)$
↘ into dk

Indeterminate Forms and L'Hospital's Rule

Example $x \rightarrow \infty: (\ln x)^n < x^p < b^x$ ($n > 0, p > 0, b > 1$)

Find $\lim_{x \rightarrow \infty} \frac{(\ln x)^n}{x} = 0$

$$= \lim_{x \rightarrow \infty} \frac{n(\ln x)^{n-1} \cdot \frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{n(\ln x)^{n-1}}{x}$$
$$= \lim_{x \rightarrow \infty} \frac{n(n-1)(\ln x)^{n-2}}{x}$$

$\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$

$\lim_{x \rightarrow \infty} \frac{x^3}{b^x} = 0 \rightarrow$ same idea of always repeating L'Hopital's rule

Solution:

$$= \lim_{x \rightarrow \infty} \frac{n(n-1)(n-2) \dots \ln x}{x} = 0$$

We have

$$\lim_{x \rightarrow 1} \ln x = \ln 1 = 0$$

$$\lim_{x \rightarrow 1} (x-1) = 0$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x-1} = 0$$

Therefore, the limit is an indeterminate form of type $\frac{0}{0}$.

We can apply L'Hospital's Rule:

$$\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)} = \lim_{x \rightarrow 1} \frac{1/x}{1} = \lim_{x \rightarrow 1} \frac{1}{x} = 1$$

Indeterminate Forms and L'Hospital's Rule

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ (or $-\infty$), then it isn't clear what the value of $\lim_{x \rightarrow a} [f(x)g(x)]$, if any, will be.

This kind of limit is called an **indeterminate form of type $0 \cdot \infty$** .

We can deal with it by writing the product fg as a quotient:

$$fg = \frac{f}{1/g} \text{ or } fg = \frac{g}{1/f}$$

This converts the given limit into an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, so that we can use L'Hospital's Rule.

Indeterminate Forms and L'Hospital's Rule

Example let $x = \frac{1}{t}$

Evaluate $\lim_{t \rightarrow \infty} \frac{1}{t} \ln\left(\frac{1}{t}\right)$

differentiate $= \lim_{t \rightarrow \infty} \frac{-\ln t}{t}$

$\lim_{x \rightarrow 0^+} x \ln x$
 $\nearrow 0$
 $\searrow -\infty$

$\lim_{x \rightarrow 0^+} \frac{\ln x \rightarrow -\infty}{\frac{1}{x} \rightarrow \infty}$

more complicated $\hookrightarrow \lim_{x \rightarrow 0^+} \frac{x}{(\ln x)^{-1}} = \lim_{x \rightarrow 0^+} \frac{x}{(-1)(\ln x)^{-2}}$

Solution: $= \lim_{t \rightarrow \infty} \frac{-\frac{1}{t}}{1} = 0$

The given limit is indeterminate. Because as $x \rightarrow 0^+$, the first factor ($x \rightarrow 0^+$), while the second factor ($\ln x \rightarrow \infty$).

Since $x = \frac{1}{(1/x)}$, we have $1/x \rightarrow \infty$ as $x \rightarrow 0^+$. Use L'Hospital's Rule

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\overset{-\infty}{\ln x}}{\underset{\infty}{1/x}} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

Note: Another possible way is to write $\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{x}{1/\ln x}$. However, if we apply L'Hospital's Rule, we get a more complicated expression than the one we started with.

In general, when we rewrite an indeterminate product, we try to choose the option that leads to the simpler limit.

Indeterminate Forms and L'Hospital's Rule

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$, then the limit

$$\lim_{x \rightarrow a} [f(x) - g(x)]$$

is called an **indeterminate form of type $\infty - \infty$** . Again there is a contest between $f(x)$ and $g(x)$.

We try to convert the difference into a quotient (for instance, by using a common denominator, or rationalization, or factoring out a common factor), so that we have an indeterminate form of type $\frac{0}{0}$ or ∞/∞ .

Indeterminate Forms and L'Hospital's Rule

Example

Compute

$$\lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x)$$

Solution:

Since $\sec x \rightarrow \infty$, $\tan x \rightarrow \infty$ as $x \rightarrow (\pi/2)^-$, the limit is indeterminate.

Use a common denominator,

$$\begin{aligned} \lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x) &= \lim_{x \rightarrow (\pi/2)^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) \\ &= \lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x} = \lim_{x \rightarrow (\pi/2)^-} \frac{-\cos x}{-\sin x} = 0 \end{aligned}$$

Note that the use of L'Hospital's Rule is justified, because $1 - \sin x \rightarrow 0$ and $\cos x \rightarrow 0$ as $x \rightarrow (\pi/2)^-$.

Indeterminate Forms and L'Hospital's Rule

Several indeterminate forms arise from the limit

$$\lim_{x \rightarrow a} [f(x)]^{g(x)}$$

- ① $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ type 0^0
- ② $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = 0$ type ∞^0
- ③ $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$ type 1^∞

These three cases can be treated either by taking the natural logarithm:

$$y = [f(x)]^{g(x)} \Rightarrow \ln y = g(x) \ln f(x)$$

or by writing the function as an exponential:

$$y = [f(x)]^{g(x)} \Rightarrow y = e^{\ln y} = e^{g(x) \ln f(x)}$$

In either method, we have the indeterminate product $g(x) \ln f(x)$, which is of type $0 \cdot \infty$.

Indeterminate Forms and L'Hospital's Rule

Example

Calculate

$$\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x}.$$

Solution:

As $x \rightarrow 0^+$, we have $1 + \sin 4x \rightarrow 1$ and $\cot x \rightarrow \infty$, so the given limit is indeterminate (type 1^∞). Let $y = (1 + \sin 4x)^{\cot x}$, we have

$$\ln y = \cot x \ln(1 + \sin 4x) = \frac{\ln(1 + \sin 4x)}{\tan x}.$$

Use L'Hospital's Rule,

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 4x)}{\tan x} \xrightarrow{\frac{0}{0} \text{ form}} \lim_{x \rightarrow 0^+} \frac{\frac{4 \cos 4x}{1 + \sin 4x}}{\sec^2 x} = 4.$$

To find limit of y , we use the fact that $y = e^{\ln y}$:

$$\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x} = \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\ln y} = e^4$$

Indeterminate Forms and L'Hospital's Rule

Cauchy's Mean Value Theorem $\rightarrow$ generalisation of MVT

Suppose that the function f and g are continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) \neq 0$ for all x in (a, b) .

Then there is a number of c in (a, b) such that

if
 $g(x) = x$

$g'(x) = 1$

$\rightarrow$ will go back to MVT

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

MVT:

1. $f(x)$ is cont $[a, b]$

2. $f(x)$ is diff (a, b)

$\exists c \in (a, b)$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Prove:

We can define a new function

$$h(x) = \overset{\text{cont}}{f(x)} - \overset{\text{constant}}{f(a)} - \frac{\overset{\text{constant}}{f(b) - f(a)}}{\overset{\text{cont}}{g(b) - g(a)}} (\overset{\text{cont}}{g(x)} - \overset{\text{constant}}{g(a)})$$

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a} (x - a)$$

Rolle's Theorem

same 2 conditions as MVT

3. $f(a) = f(b)$
 $f'(c) = 0$

It can be seen that $h(x)$ is continuous on $[a, b]$, differentiable on (a, b) , and $h(a) = h(b)$. From Rolle's theorem, we have a number $c \in (a, b)$,

$$h'(c) = 0 \Rightarrow f'(c) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(c) = 0 \Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$